FYUGP B.Sc. PHYSICS HONOURS

PROPERTIES OF SOLIDS

Semester V · Major Elective (Materials Science)

MODULE IV: DIELECTRIC AND MAGNETIC PROPERTIES OF SOLIDS

Past questions

Section A: 1 to 3 weightage

Types of magnetism

1 Distinguish between paramagnetic and ferromagnetic materials. **(1 weightage)** *November 2019 (M.Sc.)*

2 If the magnetization and flux density of a magnetic material be 3200 A/m and 0.005 Wb/m^2. Calculate the relative permeability of the material. **(3 weightage)** *November 2022 (M.Sc.)*

3 Distinguish between para and ferromagnetism. **(1 weightage)** *November 2023 (M.Sc.)*

Diamagnetism, paramagnetism and ferromagnetism (classical theories)

4 Calculate the value of magnetic susceptibility for a paramagnetic material (N = 9×10^{28} m^{-3}, $\mu_B = 9.27 \times 10^{-24}$ JT^{-1}) at 0.1 K. **(3 weightage)** *November 2019 (M.Sc.)*

5 Explain the origin of the diamagnetism. **(1 weightage)** *November 2021 (M.Sc.)*

6 A paramagnetic material has 10^{28} atoms/m^3. Its susceptibility at 350 K is 2.8×10^{-4}. Calculate susceptibility at 300 K. **(3 weightage)** *November 2021 (M.Sc.)*

7 Distinguish between hard and soft magnetic materials. **(1 weightage)** *November 2022 (M.Sc.)*

8 A ferromagnetic material with J = 3/2 and g = 2 has a Curie temperature of 125 K. Calculate the intrinsic flux density near 0 K. Also, calculate the ratio of the magnetization at 300 K in the presence of an external field of 1 mT to the spontaneous magnetization at 0 K. **(3 weightage)** *November 2023 (M.Sc.)*

Antiferromagnetism and ferrimagnetism

9 Briefly explain the origin of ferrimagnetic moments. **(1 weightage)** *November 2019 (M.Sc.)*

10 Explain Neel's theory of anti-ferromagnetism. **(1 weightage)** *November 2022 (M.Sc.)*

Polarisation, local field, dielectric constant and polarisability

11 Helium gas contains 2.7×10^{25} atoms/m^3 and dielectric constant of He atom NTD is 1. 0000684. Calculate the electric polarizability of He atom. **(3 weightage)** *November 2021 (M.Sc.)*

12 Derive the Clausius-Mossotti relation by considering the local field effects. **(3 weightage)** *November 2023 (M.Sc.)*

Ferro- and piezoelectricity

13 Distinguish between pyroelectric and piezoelectric materials. **(1 weightage)** *November 2021 (M.Sc.)*

14 Discuss the ferroelectric and paraelectric states. **(1 weightage)** *November 2023 (M.Sc.)*

Section B: 4 to 6 weightage

Diamagnetism, paramagnetism and ferromagnetism (classical theories)

15 Derive Curie-Weiss law for ferromagnetism. **(6 weightage)** *November 2019 (M.Sc.)*

16 Describe the Langevin's theory of paramagnetism and obtain an expression for paramagnetic susceptibility. Mention the temperature dependence of susceptibility. **(5 weightage)** *November 2023 (M.Sc.)*

Antiferromagnetism and ferrimagnetism

17 Distinguish between ferromagnetism and anti-ferromagnetisms. Describe the Neel model of antiferromagnetism. **(5 weightage)** *November 2021 (M.Sc.)*

Polarisation, local field, dielectric constant and polarisability

18 Describe macroscopically the origin of static dielectric constant of a substance and discuss the classical theory of electronic polarization. **(5 weightage)** *November 2022 (M.Sc.)*